

Topic: Diamond Price Prediction

DSC 680

Project #2

Milestone 2 - Whitepaper

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Introduction

This project applies supervised machine learning regression methods to predict diamond prices using the well-known Diamonds Dataset. The goal is to model the relationship between the physical and qualitative characteristics of diamonds, such as carat, cut, color, and clarity, and their corresponding market prices. By combining statistical analysis, feature engineering, and advanced regression algorithms, the study identifies the most influential factors affecting price and evaluates which model provides the most accurate predictions. These insights can help jewelers, appraisers, and online retailers develop data-driven pricing strategies that improve market transparency and customer trust.

Business Problem

The global diamond industry depends heavily on accurate valuation, yet traditional appraisal methods often rely on human expertise and subjective judgment. Price inconsistencies can occur because of different interpretations of quality grades or changing market conditions. This project explores how machine learning models can improve pricing accuracy by analyzing historical diamond data and learning patterns from multiple features. A reliable predictive model can help standardize valuation processes, improve customer satisfaction, and support strategic pricing decisions for diamond retailers and investors.

Background and History

Diamonds are valued based on the “Four Cs”: Carat, Cut, Color, and Clarity, as defined by the Gemological Institute of America (GIA). These features form the foundation of diamond grading and pricing. Historically, appraisers have manually assessed diamonds under magnification and lighting, which introduces variation and bias. With the rise of e-commerce and digital appraisals, the jewelry industry increasingly relies on data analytics for consistent, objective, and scalable pricing systems. Predictive modeling offers a solution by quantifying how each characteristic influences price and allowing accurate estimates before physical appraisal.

Data Explanation

The dataset, sourced from Kaggle, contains over 53,000 diamond entries with attributes describing both physical measurements and qualitative grades.

Feature summary:

- **Carat:** Weight of the diamond (0.2–5.01).
- **Cut:** Grading of cut quality (Fair to Ideal).
- **Color:** Grading from D (best) to J (worst).

- **Clarity:** Levels of internal flaws (I1 to IF).
- **x, y, z:** Physical dimensions (length, width, depth in mm).
- **Price:** Target variable, ranging from \$326 to \$18,823.

Data preparation steps:

1. Removed duplicates and missing entries.
2. Dropped records with zero or invalid measurements.
3. Encoded categorical variables (cut, color, clarity) using Label Encoding.
4. Scaled numerical features using StandardScaler.
5. Created an engineered feature, $\text{volume} = x \times y \times z$, and applied a log transformation on price to reduce skew.

Methods

Analysis was conducted in Python using pandas, scikit-learn, seaborn, matplotlib, and XGBoost. The following regression models were trained and evaluated:

1. **Linear Regression** – baseline model.
2. **Random Forest Regressor** – ensemble averaging model.
3. **Gradient Boosting Regressor** – sequential boosting model.
4. **XGBoost Regressor** – optimized gradient boosting implementation with hyperparameter tuning.

Model evaluation metrics included Mean Squared Error (MSE), Mean Absolute Error (MAE), R^2 score, and custom accuracy (percentage of predictions within 10% of the true price).

Hyperparameter tuning for XGBoost was performed using RandomizedSearchCV with parameters:

$n_estimators = [100-300]$, $max_depth = [3-7]$, $learning_rate = [0.01-0.1]$, $subsample = [0.7-1.0]$.

Analysis

Feature engineering revealed that carat and volume had the strongest positive correlation with price, while cut, color, and clarity showed smaller but still important effects.

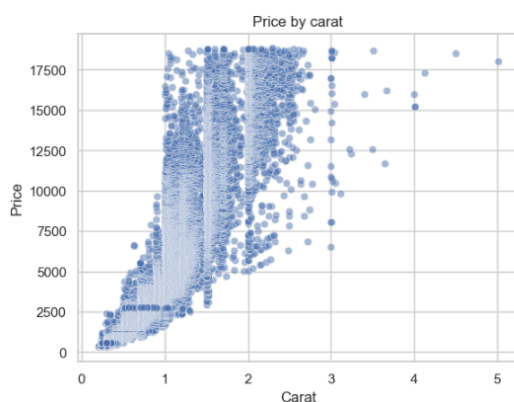


Figure 1: Scatter plot of Carat vs. Price (shows strong positive trend)

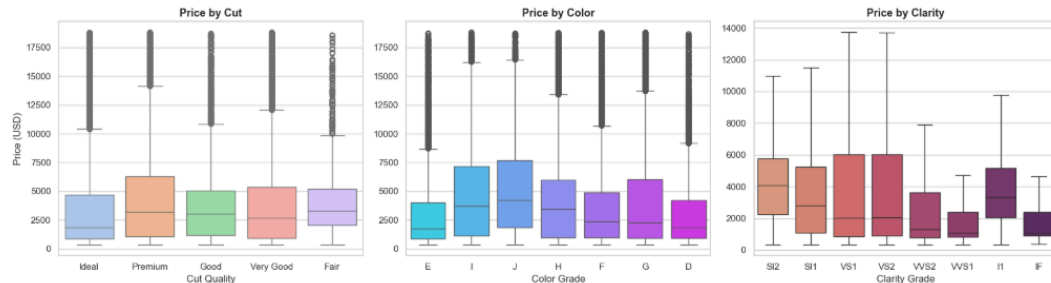


Figure 2: Box plots of Price vs. Cut, Color, and Clarity

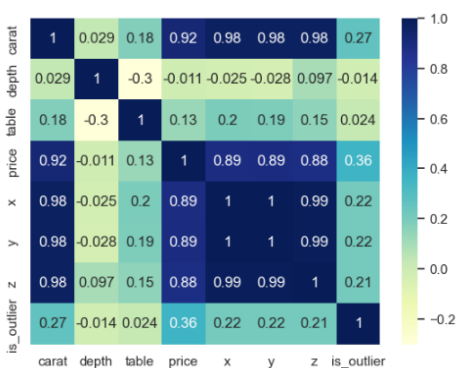


Figure 3: Correlation heatmap highlighting carat and volume as dominant predictors

Outlier removal improved data balance and reduced distortion from extremely large or poorly measured diamonds. Model performance comparison showed:

Model	MSE	MAE	R ²	Accuracy (±10%)
Linear Regression	1.70e+06	830	0.891	24.8%
Random Forest	2.55e+05	258	0.984	79.9%
Gradient Boosting	2.53e+05	276	0.984	68.6%
XGBoost	2.57e+05	277	0.985	82.0%

The XGBoost Regressor achieved the best overall performance, demonstrating strong predictive power and minimal overfitting.

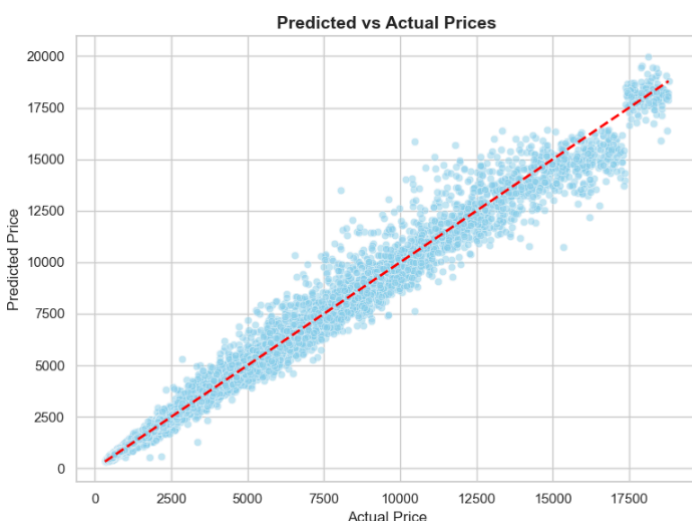


Figure 4: Predicted vs. Actual Price scatter plot showing close alignment along the diagonal.

Conclusion

This study demonstrates that machine learning can effectively predict diamond prices with high accuracy. Among all tested models, XGBoost achieved the best results, with an R² of 0.985 and accuracy of 82% within a ±10% range. These findings confirm that the four Cs, combined with physical measurements, provide a reliable basis for price prediction. Such models can support pricing automation, improve transparency, and streamline valuation in jewelry retail and e-commerce settings.

Assumptions

1. The dataset accurately represents the diamond market.
2. Historical relationships between features and price remain stable.
3. Label encoding preserves ordinal meaning for categorical features.
4. The log transformation improves model linearity without biasing predictions.

Limitations

The dataset does not account for market trends, brand effects, or certification differences that may influence price. Some high-end attributes such as fluorescence and symmetry were not included. The model is limited to the price range observed in the dataset and may not generalize well to rare or fancy-colored diamonds outside that range.

Challenges

Challenges included handling data skew, removing invalid entries, and balancing interpretability with predictive performance. Hyperparameter tuning also required computational resources to optimize XGBoost parameters efficiently. Maintaining interpretability in ensemble models was another challenge because feature importance scores do not fully explain non-linear relationships between features.

Future Uses and Additional Applications

The model can be expanded to include real-time market pricing, online appraisal systems, and integration with retail APIs for dynamic price recommendations. Future work could explore deep learning or multi-task regression to predict both price and grading simultaneously. Further studies might also test transfer learning using auction and resale data to understand depreciation patterns and collector valuations.

Recommendations

1. Adopt XGBoost for automated price estimation in digital appraisal tools.
2. Regularly retrain models with new market data to maintain accuracy.
3. Include additional features such as fluorescence and symmetry to improve prediction.
4. Use explainable AI tools like SHAP to interpret model predictions for business users.

Implementation Plan

Implementation involves developing a web-based prediction interface where users input diamond attributes and receive estimated prices. A pipeline using StandardScaler and the trained XGBoost model can run predictions in real time. Regular data updates and retraining cycles every six months are recommended to maintain accuracy as market conditions change.

Ethical Assessment

The predictive model should be used responsibly to enhance transparency, not manipulate consumer behavior or pricing. Ethical use requires clear disclosure that machine learning estimates complement professional gemological evaluations rather than replace them. The dataset must also be used in compliance with licensing agreements and without bias toward specific diamond types or regions.

10 Audience Questions

1. Why was XGBoost chosen over Random Forest despite similar R^2 scores?
2. How does log transformation improve prediction accuracy?
3. Could the model handle colored or lab-grown diamonds?
4. How were categorical features (cut, color, clarity) encoded?
5. What does 82% accuracy within 10% tolerance mean in practice?
6. How might new features like polish or fluorescence affect results?
7. What risks exist when using automated appraisal tools online?
8. Could this model be integrated into online diamond marketplaces?
9. How can interpretability be improved for non-technical users?
10. What ethical safeguards should companies adopt when deploying this model?

References

Kaggle. (2025). *Diamonds Dataset*. Retrieved from <https://www.kaggle.com/datasets/shivam2503/diamonds>

Gemological Institute of America (GIA). (2024). *Diamond 4Cs: The Methodology of Grading and Valuation*.